# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN A SERIES OF ADDS, MULTIPLIES AND DIVIDES
DATE JUNE 2, 1958 (32 OF EACH) ONE SET OF BITS PER WORD

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___
NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ✓

MAIN MEMORY CYCLE [1, 2, 2, 1,] 1, 2, 2, 1,
FAST MEMORY CYCLE [1, 2,] 1, 2,

HIGH SPEED EXCHANGE RATE NONE
MEDIUM SPEED EXCHANGE RATE NONE

OTHER INITIAL DELAY 2.3 μsecs

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED X704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 6 | 1.4 μsec | 61.1 | 58.8 | 1.84 | (13.04X) (169.6X) (USING PROG.) |
| | 8 | 1.4 | 61.1 | 58.8 | 1.84 | |
| | 16 | 1.8 | 66.4 | 64.1 | 2.00 | |
| | 32 | 2.7 | 93.7 | 91.4 | 2.86 | (8.39X) (Full wo. 704) (USING PROG.) |
| | 48 | 3.6 | 122.5 | 120.2 | 3.76 | |
| | 64 | 4.5 | 151.3 | 149.0 | 4.66 | (72.3X) |
| MPY | 6 | 8.6 | 282.5 | 280.2 | 8.76 | (27.40X) (68.5X PROG) |
| | 8 | 9.5 | 311.3 | 309.0 | 9.66 | |
| | 16 | 11.7 | 381.7 | 379.4 | 11.86 | |
| | 32 | 16.2 | 525.7 | 523.4 | 16.36 | 14.67X |
| | 48 | 20.7 | 669.7 | 667.4 | 20.86 | (Prog. 65.2X) |
| DIV | 6 | 10.8 | 352.9 | 350.6 | 10.96 | (21.90X) (56.9X Prog) |
| | 8 | 11.3 | 365.9 | 363.6 | 11.46 | |
| | 16 | 13.1 | 426.5 | 424.2 | 13.26 | |
| | 32 | 16.7 | 542.3 | 540.0 | 16.86 | 14.23X |
| | 48 | 20.7 | 661.7 | 659.4 | 20.86 | |

COMMENTS RUN NO. 1.

# PROJECT 7090 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN SERIES OF ADD, MPY, DIV, ALL CROSSING WD. BOUNDARY (WORST CASE OF CONFLICTS)

DATE 3 JUNE 1958

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ✓ BOTH ___

MAIN MEMORY CYCLE [1, 2, 2, 1, 2, 1, 1, 2,] 1, 2, 2, 1, 2, 1, 1,

FAST MEMORY CYCLE [1, 2,] 1, 2,

HIGH SPEED EXCHANGE RATE NONE

MEDIUM SPEED EXCHANGE RATE NONE

OTHER INITIAL DELAY 2.3 μSECS

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED X704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 6 | 1.4 μsecs | 109.7 | 107.4 | 3.36 | |
| | 8 | 1.4 | 109.7 | 107.4 | 3.36 | |
| | 16 | 1.8 | 119.1 | 116.8 | 3.65 | |
| | 32 | 2.7 | 147.9 | 145.6 | 4.55 | |
| | 48 | 3.6 | 176.7 | 174.4 | 5.45 | |
| | 64 | 4.5 | 205.5 | 203.2 | 6.35 | |
| MPY | | | | | | |
| | 6 | 8.6 | 330.5 | 328.2 | 10.25 | |
| | 8 | 9.5 | 359.3 | 357.0 | 11.15 | |
| | 16 | 11.7 | 435.9 | 433.6 | 13.50 | |
| | 32 | 16.2 | 579.9 | 577.6 | 18.00 | |
| | 48 | 20.7 | 723.9 | 721.6 | 22.50 | |
| DIV | | | | | | |
| | 6 | 10.8 | 407.1 | 404.8 | 12.64 | |
| | 8 | 11.3 | 416.9 | 414.6 | 12.95 | |
| | 16 | 13.1 | 474.5 | 472.2 | 14.78 | |
| | 32 | 16.7 | 588.3 | 586.0 | 18.35 | |
| | 48 | 20.7 | 723.9 | 721.6 | 20.50 | |

COMMENTS RUN NO. 2

# PROJECT 7090 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN 6 BITS CROSSING WD. BOUNDARY (ADDS)
DATE JUNE 5, 1958 WITH VARIOUS COMBINATIONS OF HIGH AND MEDIUM SPEED EXCHANGES.

Assumptions

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ___ BOTH ___
NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ___ ✓ BOTH ___
MAIN MEMORY CYCLE [1, 2, 2, 1, 2, 1, 1, 2]
FAST MEMORY CYCLE [1, 2, 7]

HIGH SPEED EXCHANGE RATE ∞; 7.1; 7.1; 7.1; 7.1; 7.1; 7.1;
MEDIUM SPEED EXCHANGE RATE ∞; ∞; 32; 16; 8; 4; 2;

OTHER INITIAL DECAY 2.3 μsecs

| TYPE OF ADD HI LO | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | ~~SPEED X'CH RATE~~ MEMORY CONFLICTS |
|---|---|---|---|---|---|---|
| ∞ ∞ | 6 | 1.4 μsecs | 109.7 μsecs | 107.4 μsecs | 3.36 μsecs | 8.7% |
| 7.1 ∞ | " | " | 151.1 | 148.8 | 4.73 | 48.1 |
| 7.1 32 | " | " | 165.8 | 163.5 | 5.11 | 66.2 |
| 7.1 16 | " | " | 179.6 | 177.3 | 5.55 | 76.9 |
| 7.1 8 | " | " | 221.3 | 219.0 | 6.84 | 111.7 |
| 7.1 4 | " | " | 440.0 | 437.7 | 13.70 | 179.9 |
| 7.1 2 | " | " | STOPPED UP | | | |

COMMENTS RUN NO. 3

Theoretical Max. rate for Med. Speed Exchange + High Speed Combined $\frac{1}{X} = \frac{1}{2.1} - \frac{1}{7.1}$ $X = 2.98$ μsec per Read-Write for 2MM.

2.1 ← mem cycle; 7.1 ← High Speed Exchange Rate

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS. ALL CROSSING WORD BOUNDARIES
DATE JUNE 6, 1958 WITH VARIOUS COMBINATIONS OF HIGH AND MED SPEED EXCHANGES

ASSUMPTIONS

NO. OF MAIN MEMORIES 4 DATA ONLY ✓ INSTRUCTION ONLY — BOTH —
NO. OF FAST MEMORIES 2 DATA ONLY — INSTRUCTION ONLY ✓ BOTH —
MAIN MEMORY CYCLE [1, 2, 2, 3, 2, 3, 1, 2, 3, 4, 4, 1, 4, 1, 3, 4,]
FAST MEMORY CYCLE [1, 2,] 1 2,

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE
OTHER INLOOK DELAY 2.3

| TYPE OP. HEX | LEX. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 | REMARKS |
|---|---|---|---|---|---|---|---|---|
| ∞ | ∞ | DIGIT | 2.7 | 1049 | 102.6 | 3.20 | | |
| 7.1 | ∞ | DIGIT | 2.7 | 126.5 | 124.2 | 3.89 | | |
| 7.1 | 32 | " | 2.7 | 122.0 | 119.7 | 3.74 | | |
| 7.1 | 16 | | 2.7 | 114.8 | 112.5 | 3.52 | | |
| 7.1 | 8 | | 2.7 | 136.7 | 134.4 | 4.20 | | |
| 7.1 | 4 | | | 179.6 | 177.3 | 5.53 | | |
| 7.1 | 2 | | | 215.6 | 213.3 | 6.67 | | |
| 7.1 | 1.5 | | | 291.5 | 289.2 | 9.01 | | |

COMMENTS RUN NO. 4

Theoretical Max. Rate for 4 MM's

$$\frac{1}{x} = \frac{2}{2.1} - \frac{1}{7.1}$$

$x = 0.81$ μs per word-minute

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN ADDS, MPYS, DIVS, ALL CROSSING WD. BOUNDARIES

DATE JUNE 6, 58

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY ✓ BOTH __

NO. OF FAST MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY __ BOTH __

MAIN MEMORY CYCLE [1, 2,]

FAST MEMORY CYCLE [1, 2, 2, 1, 2, 1, 1, 2,]

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER INDEX ARRAY 2.7

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 6 | 1.4 | 90.7 | 88.0 | 2.75 | |
| | 8 | 1.4 | 90.7 | 88.0 | 2.75 | |
| | 16 | 1.8 | 100.4 | 97.7 | 3.05 | |
| | 32 | 2.7 | 129.2 | 126.5 | 3.96 | |
| | 48 | 3.6 | 158.0 | 155.3 | 4.89 | |
| | 64 | 4.5 | 186.8 | 184.1 | 5.75 | |
| | | | | | | |
| MPY | 6 | 8.6 | 321.1 | 318.4 | 9.96 | |
| | 8 | 9.5 | 349.9 | 347.2 | 10.84 | |
| | 16 | 11.7 | 417.2 | 414.5 | 12.95 | |
| | 32 | 16.2 | 561.2 | 558.5 | 17.40 | |
| | 48 | 20.7 | 705.2 | 702.5 | 21.90 | |
| | | | | | | |
| DIV | 6 | 10.8 | 388.4 | 385.7 | 12.04 | |
| | 8 | 11.3 | 407.5 | 404.8 | 12.65 | |
| | 16 | 13.1 | 465.1 | 462.4 | 14.45 | |
| | 48 | 20.7 | 705.2 | 702.5 | 21.90 | |

COMMENTS RUN NO. 5.

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN ADDS, MPYS, DIVS, NO CROSSING WORD BOUNDARIES

DATE JUNE 6, 1958

ASSUMPTIONS

NO. OF MAIN MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __

NO. OF FAST MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

MAIN MEMORY CYCLE { INST [1, 2, 1, 2,

FAST MEMORY CYCLE { DATA [1, 2, 2, 1, 2, 1, 1, 2,

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER DELAY 2,3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | | | | | | |
| | 6 | 1.4 | 99.7 | 97.4 | 3.05 | |
| | 8 | 1.4 | 99.7 | 97.4 | 3.05 | |
| | 16 | 1.8 | 104.6 | 102.3 | 3.20 | |
| | 32 | 2.7 | 128.6 | 126.3 | 3.95 | |
| | 64 | 4.5 | 186.2 | 183.9 | 5.74 | |
| | | | | | | |
| MPY | 6 | 8.6 | 320.5 | 318.2 | 9.95 | |
| | 16 | 11.7 | 416.6 | 414.3 | 12.90 | |
| | 48 | 20.7 | 704.6 | 702.3 | 22.00 | |
| | | | | | | |
| DIV | 16 | 10.8 | 387.8 | 385.5 | 12.00 | |
| | 16 | 13.1 | 469.5 | 462.2 | 14.40 | |
| | 48 | 20.7 | 704.6 | 702.3 | 22.00 | |

COMMENTS RUN NO. 6

NTB 4.2-58

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN ADDS, MPYS, DIVS, ALL CROSSING WD. BOUNDARIES

DATE JUNE 6, 58

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

NO. OF FAST MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __

MAIN MEMORY CYCLE / FAST MEMORY CYCLE
INST [1, 2, 1, 2,]
DATA [1, 2, 2, 1, 2, 1, 1, 2,]

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER INITIAL DELAY 2.7

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 6 | 1.4 | 134.0 | 131.3 | 4.10 | |
| | 8 | 1.4 | 134.0 | 131.3 | 4.10 | |
| | 16 | 1.8 | 138.6 | 135.9 | 4.25 | |
| | 32 | 2.7 | 153.0 | 150.3 | 4.70 | |
| | 64 | 4.5 | 205.8 | 203.1 | 6.35 | |
| | | | | | | |
| MPY | 6 | 8.6 | 330.8 | 328.1 | 10.30 | |
| | 16 | 11.7 | 436.2 | 433.5 | 13.60 | |
| | 48 | 20.7 | 724.2 | 721.5 | 22.50 | |
| | | | | | | |
| DIV | 6 | 10.8 | 407.4 | 404.7 | 12.60 | |
| | 16 | 13.1 | 474.8 | 472.1 | 14.80 | |
| | 48 | 20.7 | 724.2 | 721.5 | 22.50 | |

COMMENTS RUN NO. 7

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS, ALL CROSSING WD. BOUNDARIES
DATE JUNE 6 '58 WITH BOTH EXCHANGES

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓
NO. OF FAST MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __
MAIN MEMORY CYCLE { HIGH + MED EXCG. READ [1, 2] } { ~~HIGH + MED EXCG. WRITE [2, 1]~~ }
FAST MEMORY CYCLE

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE
OTHER DELAY 2.7

| TYPE OP. ~~HOP LOP~~ | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| 20 00 | | | 134.0 | 131.3 | 4.10 | |
| 7.1 00 | | | 180.2 | 177.5 | 5.55 | |
| 7.1 32 | | | 194.6 | 191.9 | 6.00 | |
| 7.1 16 | | | 221.3 | 218.6 | 6.83 | |
| 7.1 8 | | | 269.3 | 266.6 | 8.35 | |
| 7.1 4 | | | 547.4 | 544.7 | 17.00 | |

COMMENTS RUN NO. 8

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS, ALL CROSSING WORD BOUNDARIES

DATE JUNE 6, 58 WITH BOTH EXCHANGES

ASSUMPTIONS

NO. OF MAIN MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __

NO. OF FAST MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

MAIN MEMORY CYCLE

FAST MEMORY CYCLE (SAME AS RUN 8)

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER DELAY 2.3

| TYPE OP. ~~NOT~~ ~~EFF~~ | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ∞ ∞ | | | 99.7 | 97.4 | 3.04 | |
| 7.1 ∞, | | | 108.7 | 106.4 | 3.34 | |
| " 32 | | | 107.8 | 105.5 | 3.30 | |
| 16 | | | 105.7 | 103.4 | 3.25 | |
| 8 | | | 115.0 | 112.7 | 3.52 | |
| 4 | | | 122.8 | 120.5 | 3.77 | |
| 2 | | | 141.4 | 139.1 | 4.35 | |
| 1.5 | | | 164.2 | 161.9 | 5.07 | |

COMMENTS RUN NO. 9

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN ADD, MPY & DIV ALL FROM SEQUENTIAL LOADS

DATE JUNE 6'58

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ✓ BOTH ___
NO. OF FAST MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___

MAIN MEMORY CYCLE [1, 2, 1]
FAST MEMORY CYCLE [1, 2, 2, 1]

HIGH SPEED EXCHANGE RATE NONE
MEDIUM SPEED EXCHANGE RATE NONE

OTHER DELAY 2.7

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 16 | 1.8 | 64.8 | 62.1 | 1.94 | |
| | 32 | 2.7 | 93.6 | 90.9 | 2.84 | |
| | 64 | 4.5 | 151.2 | 148.5 | 4.64 | |
| | 6-8 | 1.4 | 52.0 | 49.3 | 1.54 | |
| MPY | 6 | 8.6 | 282.4 | 279.7 | 8.74 | |
| | 16 | 11.7 | 381.6 | 378.9 | 11.84 | |
| | 48 | 20.7 | 669.6 | 666.9 | 20.84 | |
| | | | | | | |
| DIV | 6 | 10.8 | 352.8 | 350.1 | 10.94 | |
| | 16 | 13.1 | 426.4 | 423.7 | 13.24 | |

COMMENTS RUN NO. 10

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN ADDS, MPYS, DIVS SEPARATE WORDS

DATE JUNE 6 '58

ASSUMPTIONS

NO. OF MAIN MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __
NO. OF FAST MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

MAIN MEMORY CYCLE { INST [1, 2, 1, 2,]
FAST MEMORY CYCLE { DATA [1, 2, 2, 1]

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE

OTHER DECIMAL 2.3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 16 | 1.8 | 64.5 | 62.7 | 1.91 | |
| | 32 | 2.7 | 93.3 | 91.0 | 2.84 | |
| | 64 | 4.5 | 150.9 | 148.6 | 4.65 | |
| | 6-8 | 1.4 | 52.0 | 49.3 | 1.54 | |
| MPY | 6 | 8.6 | 282.1 | 279.8 | 8.74 | |
| | 16 | 11.7 | 381.3 | 379.0 | 11.82 | |
| | 48 | 20.7 | 669.3 | 667.0 | 20.85 | |
| | | | | | | |
| DIV | 6 | 10.8 | 352.5 | 352.2 | 11.00 | |
| | 16 | 13.1 | 426.1 | 423.8 | 13.20 | |

COMMENTS RUN NO. 11

NJB 6-2-58

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN ADDS, MPY, DIV, STRAIGHT MIXED

DATE JUNE 5

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ✓
NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

MAIN MEMORY CYCLE { SAME AS RUN 11 }
FAST MEMORY CYCLE

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE

OTHER DECAL 2.7

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ADD | 12 | 1.8 | 81.1 | 78.4 | 2.45 | |
| | 30 | 2.7 | 95.5 | 92.8 | 2.90 | |
| | 60 | 4.5 | 151.6 | 148.9 | 4.65 | |
| | 6-8 | 1.4 | 78.2 | 75.5 | 2.36 | |
| MPY | 6 | 8.6 | 282.8 | 280.1 | 8.75 | |
| | 16 | 11.7 | 382.0 | 379.3 | 11.85 | |
| | 48 | 20.7 | 670.0 | 667.3 | 20.85 | |
| | | | | | | |
| DIV | 6 | 10.8 | 353.2 | 350.5 | 10.95 | |
| | 16 | 13.1 | 426.8 | 424.1 | 13.25 | |

COMMENTS RUN NO 12.

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS, SCRAMBLE MODES

DATE JUNE 6 58

ASSUMPTIONS

NO. OF MAIN MEMORIES 8 DATA ONLY ✓ INSTRUCTION ONLY — BOTH —

NO. OF FAST MEMORIES — DATA ONLY — INSTRUCTION ONLY ✓ BOTH —

MAIN MEMORY CYCLE [1, 2, 2, 1]

FAST MEMORY CYCLE [1, 2]

HIGH SPEED EXCHANGE RATE 7.1

MEDIUM SPEED EXCHANGE RATE

OTHER DELAY 2.2

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| 7 [illegible] | 6 | | 61.1 | 58.8 | 1.84 | |
| 7 0 | | | 79.4 | 77.1 | 2.41 | |
| 7 32 | | | 86.9 | 84.6 | 2.64 | |
| 7 16 | | | 95.0 | 92.7 | 2.89 | |
| 7 8 | | | 116.3 | 113.0 | 3.53 | |
| 7 4 | | | 268.1 | 265.8 | 8.28 | |

COMMENTS RUN NO 13

# PROJECT 7000 SIMULATION -- HIGHEST VERSION

DESCRIPTION OF RUN [illegible]

DATE [illegible]

ASSUMPTIONS

NO. OF MAIN MEMORIES 1 DATA ONLY __ INSTRUCTION ONLY __ BOTH __
NO. OF FAST MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

MAIN MEMORY CYCLE
FAST MEMORY CYCLE {SAME AS RUN 8}

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE

OTHER DEFN 1 2 3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| 1,1 | 32 | | 57.8 | 57.5 | 1.80 | |
| | 16 | | 1 | 60.8 | 1.90 | |
| | 1 | | 62.7 | 62.6 | 1.95 | |
| | 1 | | 1.1.5 | 69.2 | 2.16 | |

COMMENTS RUN IN 4

NTP 4-2-59

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS SEPARATE WORDS

DATE JUNE 6 '58

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY __ INSTRUCTION ONLY __ BOTH ✓

NO. OF FAST MEMORIES 0 DATA ONLY __ INSTRUCTION ONLY __ BOTH __

MAIN MEMORY CYCLE {SAME AS RUN 8}

FAST MEMORY CYCLE

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER DELAY 2.7

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 | REMARKS |
|---|---|---|---|---|---|---|---|
| 2.1 32 | 6 | | 109.2 | 126.5 | 3.95 | | |
| " 16 | | | 143.6 | 140.9 | 4.40 | | |
| 8 | | | 181.1 | 178.4 | 5.57 | | |
| 4 | | | 383.0 | 380.3 | 11.89 | | |

COMMENTS RUN NO. 15

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN 6 BIT ADDS CONSECUTIVE WITH EVERY
DATE DEC 10 11 CROSSING WD BOUND. (33)

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY — INSTRUCTION ONLY — BOTH —
NO. OF FAST MEMORIES 2 DATA ONLY — INSTRUCTION ONLY — BOTH —

MAIN MEMORY CYCLE
FAST MEMORY CYCLE

HIGH SPEED EXCHANGE RATE
MEDIUM SPEED EXCHANGE RATE

OTHER

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | DATA MEM CYCLE ~~SPEED FOR REMARKS~~ |
|---|---|---|---|---|---|---|
| 2 DMM 2 IFM | DEC 1 | 2.3 | 55.3 | 53.0 | 1.655 | [1,1,1,1,1,1,1,1,1,1,2,2-- 2. |
| 2 IMM 2 DFM | | 2.7 | 55.5 | 52.8 | 1.650 | same |
| 2 DMM 2 IMM | | 2.7 | 55.6 | 52.9 | 1.652 | |

COMMENTS RUN NO. 16

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN LOAD & STORE (16)

DATE JUNE 10

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY — BOTH —

NO. OF FAST MEMORIES 2 DATA ONLY — INSTRUCTION ONLY — BOTH —

MAIN MEMORY CYCLE [4, 2, 2, 2, 1, 1] TO A & B

FAST MEMORY CYCLE {1, 2}

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER 2.3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ~~LOAD A~~ LOAD B STORE B | — | .8 | 96.6 | 95.8 | 6.00 | |

COMMENTS RUN NO 17

NTB 4-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN LOAD ADD STORE (16)

DATE JUNE 11, '58 (.6) 1.4 (.2) = 2.2

ASSUMPTIONS

NO. OF MAIN MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

MAIN MEMORY CYCLE [1, 2, 2, 2, 1, 1, 2, 2, 2, 1, 1, 1]

FAST MEMORY CYCLE

HIGH SPEED EXCHANGE RATE

MEDIUM SPEED EXCHANGE RATE

OTHER DECAY 2.3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| LOAD A | 6 | 2.2 | 124.5 | 122.2 | 7.65 | |
| ADD B | 8 | 2.2 | 124.5 | 122.2 | 7.65 | |
| LOAD C | 16 | 2.6 | 126.0 | 123.7 | 7.72 | |
| STORE C | 32 | 3.5 | 133.9 | 131.6 | 8.22 | |
| | 64 | 5.3 | 162.7 | 160.4 | 10.00 | |

COMMENTS RUN NO. 18

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN LOAD & STORES (16)

DATE MAR 11, 58

ASSUMPTIONS

NO. OF MAIN MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___
NO. OF FAST MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

MAIN MEMORY CYCLE ___
FAST MEMORY CYCLE ___

HIGH SPEED EXCHANGE RATE ___
MEDIUM SPEED EXCHANGE RATE ___

OTHER ___

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME. | ~~SPEED~~ MEMORY CYCLE ~~RATE~~ REMARKS |
|---|---|---|---|---|---|---|
| IMEXCHANGE | 2.700 | 4/. | 69.2 | 66.5 | 4.15 | SAME AS RUN 17 |
| ACCINFM | 23 | | 68.9 | 66.6 | 4.15 | " |
| ACCINMM | 32 | | 113.7 | 111.0 | 6.98 | " |
| | | | | | | |
| REGWAB | | | | | | |
| 7.1 ∞ | 2.700 | 47 | 120.0 | 117.7 | 7.35 | SAME AS RUN 17 |
| " 32 | | | 143.7 | 141.4 | 8.85 | WITH |
| " 16 | | | 158.1 | 155.8 | 9.74 | EXCG READ 1,2,1,2, |
| " 8 | | | 214.7 | 212.4 | 13.30 | EXCG WRITE 2,1,2,1, |
| " 4 | | | 446.0 | 443.7 | 27.70 | ADDED |

COMMENTS RUN 19

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN LOAD ADD (LOAD) STORE

DATE JUNE 11, 58

ASSUMPTIONS

NO. OF MAIN MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___
NO. OF FAST MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

MAIN MEMORY CYCLE ___
FAST MEMORY CYCLE ___

HIGH SPEED EXCHANGE RATE ___
MEDIUM SPEED EXCHANGE RATE ___

OTHER ___

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | MEMORY CYCLE REMARKS |
|---|---|---|---|---|---|---|
| INTERCHANGED | 6 | | 92.0 | 89.3 | 5.58 | SAME AS RUN 18 |
| | 8 | | 90.0 | 89.3 | 5.58 | |
| | 16 | | 95.1 | 92.4 | 5.77 | |
| | 32 | | 109.5 | 106.8 | 6.67 | |
| | 64 | | 138.3 | 135.6 | 8.48 | |
| | | | | | | |
| ALL IN FM | 6 | | 95.0 | 92.7 | 5.79 | |
| | 8 | | 95.0 | 92.7 | 5.79 | |
| | 32 | | 110.4 | 108.1 | 6.76 | |
| | 64 | | 139.2 | 136.9 | 8.55 | |
| | | | | | | |
| ALL IN MM | 6 | | 159.5 | 156.8 | 9.80 | |
| | 8 | | 159.5 | 156.8 | 9.80 | |
| | 16 | | 159.5 | 156.8 | 9.80 | |
| | 32 | | 159.5 | 156.8 | 9.80 | |
| | 64 | | 163.6 | 160.9 | 10.05 | |
| | | | | | | |
| REVERSE | | | | | | SAME AS RUN 18 |
| 7.1 ∞ | | | 165.6 | 162.9 | 10.20 | WITH |
| " 32 | | | 192.3 | 189.6 | 11.85 | |
| " 16 | | | 207.0 | 204.3 | 12.77 | EXCG. READS 1 2 1 2 |
| " 8 | | | 264.7 | 262.0 | 16.40 | EXCG. WRITES 2 1 2 1 |
| " 4 | | | 593.6 | 590.9 | 36.90 | ADDED |
| | | | | | | |
| | | | | | | |

COMMENTS RUN 20

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION — HIGHEST VERSION

DESCRIPTION OF RUN LOAD ADD TO MEM (16) (LOAD, LOAD ADD STORE)

DATE 12 JUNE 58

ASSUMPTIONS

NO. OF MAIN MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

~~MAIN~~ DATA MEMORY CYCLE [1, 2, 2, 2, 4, 2, 2 . . .

~~FAST~~ INST MEMORY CYCLE [1, 2,]

HIGH SPEED EXCHANGE RATE ___

MEDIUM SPEED EXCHANGE RATE ___

OTHER ___

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| DATA IN 2MM | 6 | | 121.6 | 119.3 | 7.47 | |
| INST IN 2 FM | 8 | | 121.6 | 119.3 | 7.47 | |
| DECAY = 2.3 | 16 | | 126.5 | 124.2 | 7.77 | |
| | 32 | | 140.9 | 138.6 | 8.67 | |
| | 64 | | 169.7 | 167.4 | 10.46 | |
| | | | | | | |
| INTERCHANGE | 6 | | 94.2 | 91.5 | 5.72 | |
| | 8 | | 94.2 | 91.5 | 5.72 | |
| DECAY = 2.7 | 16 | | 99.1 | 96.4 | 6.02 | |
| | 32 | | 113.5 | 110.8 | 6.92 | |
| | 64 | | 142.3 | 139.6 | 8.73 | |
| | | | | | | |
| ALL IN FM | 6 | | 93.9 | 91.6 | 5.72 | |
| | 8 | | 93.9 | 91.6 | 5.72 | |
| DECAY = 2.3 | 16 | | 98.8 | 96.5 | 6.02 | |
| | 32 | | 113.2 | 110.9 | 6.92 | |
| | 64 | | 142.0 | 139.7 | 8.73 | |
| | | | | | | |
| ALL IN MM | 6 | | 126.7 | 124.0 | 7.75 | |
| | 8 | | 126.7 | 124.0 | 7.75 | |
| | 16 | | 129.2 | 126.5 | 7.90 | |
| DECAY = 2.7 | 32 | | 141.2 | 138.5 | 8.65 | |
| | 64 | | 170.0 | 167.3 | 10.46 | |
| | | | | | | |
| | | | | | | |

COMMENTS RUN NO 21



NJB 6-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN LOAD ADD TO MEM (16) WITH EXCHANGES

DATE 12 JUNE 58 6 BIT CASE

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ✓

DATA MAIN MEMORY CYCLE SAME AS RUN 21 WITH EXCG. READS [1, 2, 1, 2] and 1

~~FAST~~ MEMORY CYCLE EXCG WRITES [2, 1, 2, 1]

HIGH SPEED EXCHANGE RATE ___

MEDIUM SPEED EXCHANGE RATE ___

OTHER DECAY = 2.3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| ME ∠E | | | | | | |
| 7.1 ∞ | | | 148.4 | 146.1 | 9.14 | 8 |
| " 32 | | | 164.1 | 161.8 | 10.10 | |
| 16 | | | 176.3 | 174.0 | 10.90 | |
| 8 | | | 219.6 | 217.3 | 13.60 | |
| 4 | | | 459.9 | 457.6 | 28.60 | |

COMMENTS RUN NO 22

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN SAMPLE PROG (4 times through loop)

DATE 12 JUNE 58

ASSUMPTIONS

NO. OF MAIN MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES ___ DATA ONLY ___ INSTRUCTION ONLY ___ BOTH ___

DATA MAIN MEMORY CYCLE {[1,2,2][1,1,]B[1,2,2]A[1,]C[1,]B}

INST FAST MEMORY CYCLE [1,2]A

HIGH SPEED EXCHANGE RATE ___

MEDIUM SPEED EXCHANGE RATE ___

OTHER ___

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | ~~SPEED FOR~~ REMARKS |
|---|---|---|---|---|---|---|
| REGULAR | 6 | | 130.9 | 128.6 | 32.1 | A |
| DELAY=2.3 | 8 | | 130.9 | 128.6 | 32.1 | 1. LOAD i MM |
| | 16 | | 138.4 | 136.1 | 34.0 | 2. — |
| | 32 | | 153.7 | 151.4 | 37.9 | 3. ADD i MM |
| | 64 | | 188.5 | 186.2 | 46.5 | 4. — |
| | | | | | | 5. COMP MM |
| INTERCHANGED | 6 | | 130.1 | 127.4 | 31.9 | 6. — |
| | 8 | | 130.1 | 127.4 | 31.9 | 7. BR IND TO C |
| DELAY=2.7 | 16 | | 134.9 | 132.2 | 33.1 | B 8. LOAD MM |
| | 32 | | 150.5 | 147.8 | 36.9 | 9. STOR MM 1 |
| | 64 | | 186.5 | 183.8 | 46.0 | 10. INC IND FM |
| | | | | | | 11. COUNT+BR TO A |
| ALL IN FM | 6 | | 118.9 | 116.6 | 29.2 | C 12. ADD |
| DELAY=2.3 | 8 | | 118.9 | 116.6 | 29.2 | 13 — |
| | 16 | | 126.1 | 123.8 | 31.0 | 14 BR. |
| | 32 | | 144.1 | 141.8 | 35.4 | |
| | 64 | | 180.1 | 177.8 | 44.5 | |
| | | | | | | |
| ALL IN MM | 6 | | 162.5 | 159.8 | 39.9 | |
| | 8 | | 162.5 | 159.8 | 39.9 | |
| DELAY=2.7 | 16 | | 164.9 | 162.2 | 40.6 | |
| | 32 | | 170.9 | 168.2 | 42.0 | |
| | 64 | | 194.9 | 192.2 | 48.0 | |
| | | | | | | |
| | | | | | | |

COMMENTS RUN NO 23

ABACB GOES THRU A - 8 TIMES ; B - 8 TIMES ; C - 4 TIMES

# PROJECT 7000 SIMULATION — HARVEST VERSION

DESCRIPTION OF RUN SAMPLE PROG WITH EXCHANGES

DATE 23 JUNE 58 (4 TIMES THROUGH LOOP)

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ✓ BOTH ___

DATA
MAIN MEMORY CYCLE SAME AS RUN 23 EXCEPT WITH EXCH CYCLES ADD.
FAST MEMORY CYCLE READS & WRITES.

HIGH SPEED EXCHANGE RATE ___

MEDIUM SPEED EXCHANGE RATE ___

OTHER DECAY = 2.3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| THE 64 | | | | | | |
| 7.1 " 32 | | | 143.8 | 147.5 | 35.4 | |
| " 16 | | | 156.4 | 154.1 | 38.5 | |
| 8 | | | 172.0 | 169.7 | 42.4 | |
| 4 | | | 203.8 | 201.5 | 50.3 | |
| 2 | | | 403.6 | 401.3 | 100.3 | |
| | | | 322.0 | 319.7 | 79.7 | ? |

COMMENTS RUN NO 24

N.J.B. 6-2-58

# PROJECT 7000 SIMULATION – HARVEST VERSION

DESCRIPTION OF RUN VFL MATRIX MPY (10)

DATE _____

ASSUMPTIONS

NO. OF MAIN MEMORIES 2 DATA ONLY ✓ INSTRUCTION ONLY ___ BOTH ___

NO. OF FAST MEMORIES 2 DATA ONLY ___ INSTRUCTION ONLY ✓ BOTH ___

~~MAIN~~ FAST MEMORY CYCLE [1, 2, 1 1/2]

~~FAST~~ MAIN MEMORY CYCLE [1, 2, ]

HIGH SPEED EXCHANGE RATE _____

MEDIUM SPEED EXCHANGE RATE _____

OTHER DECAY = 2 3

| TYPE OP. | NO. BITS | OP. TIME | TOTAL TIME | CORR. TIME | AVE. TIME | SPEED x 704 REMARKS |
|---|---|---|---|---|---|---|
| COMP | 6 | | 193.9 | 191.6 | 19.2 | |
| MPY | 8 | | 193.9 | 191.6 | 19.2 | |
| ADD | 16 | | 229.9 | 227.6 | 22.8 | |
| STOR | 32 | | 283.9 | 281.6 | 28.2 | |
| INCR | | | | | | |
| A, C, & B | | | | | | |

COMMENTS RUN NO. 25



NJB. 6-2-58